

Mating Governance and Lineage Establishment — Decomposing B1.10 Mating as Cross-Layer Combination by Formalizing How Mating Events Are Governed Consistently Across All Three Patterns per B2.41 Birth Governance vs. Labor Structure, How Mating Lineage Is Established in Offspring Birth Records per B2.43, and How Cross-Partner Mating Follows A2.47 Authority Distribution

Derivation Note B2.49 — Phase B2, Series B

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Date: May 12, 2026

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This work derives from and formalizes the instinct/reasoning separation pattern introduced in “The Instinct/Reasoning Separation Outside the Model” (Li, April 2026), the second paper in the CKS theory series following “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems” (Li, April 2026).

Abstract

Mating in the Coordination Knowledge Substrate (CKS) pattern is a lifecycle primitive in which two or more parent cells, aspects, or Selves combine their content into an offspring under orchestration substrate governance. Paper 2 specifies three mating patterns — Union (B2.46), Selective merge (B2.47), and Lineage-preserved union (B2.48) — each with distinct combination mechanics. This note formalizes the governance framework that is common to all three: how mating events are governed, how governance acts are distinguished from labor acts per the B2.41 birth governance vs. labor structure, how mating establishes lineage in offspring birth records per B2.43, and how cross-partner mating is conditioned on A2.47 authority distribution. The governance framework is consistent across all three mating patterns; Selective merge adds one additional governance act (selection rule authoring) that the other two patterns do not require. Naming the governance framework as a standalone operational variant is architecturally necessary because the prior four B1.10 decomposition notes (B2.45–B2.48) formalize each mating mechanism individually; B2.49 formalizes what is common to all of them. B2.50 (mating verification) will close the six-note B1.10 decomposition.

1. Why mating governance and lineage establishment needs to be formalized as a standalone operational variant

The four preceding notes in the B1.10 decomposition — B2.45 (mating mechanism specification), B2.46 (Union), B2.47 (Selective merge), B2.48 (Lineage-preserved union) — formalize the three mating patterns individually: how each pattern combines parental content, what architectural

commitments each makes, and how each differs from the others. Together they cover the *what* of mating mechanics. They do not resolve, individually or collectively, the *how* of mating governance: who decides that a mating event occurs, under what structure, and how the mating event is recorded and connected to offspring lineage across all three patterns.

Without a standalone formalization of mating governance, the prior-art record contains three pattern-specific mechanism accounts and no account of the consistent governance framework they operate within. This gap matters both architecturally and strategically. Architecturally, the governance framework is not a by-product of any individual mating pattern; it is what makes the patterns collectively operable under human authority. Strategically, governance-over-combination-primitives is exactly the territory where implementations diverge: two systems may support similar combination mechanics but differ completely on whether and how humans govern the combination decision, the pattern selection, and the resulting offspring's lineage anchoring. The governance framework is the patentable surface that distinguishes CKS mating from ad-hoc or automatic component combination in conventional AI deployments.

B2.49 occupies the forty-ninth position in Phase B2, fifth of six notes decomposing B1.10. It formalizes the consistent cross-pattern governance framework and thereby completes the governance account of mating as a CKS lifecycle primitive. B2.50 will close the decomposition by formalizing mating verification — the confirmation that governance acts were properly executed and the offspring specification is authoritative before offspring birth proceeds per B1.09.

2. The architectural framework precisely stated

2.1 The governance vs. labor distinction applied to mating

B2.41 establishes that the birth governance vs. labor distinction from Paper 1 §3.3 applies to all lifecycle operations. The distinction holds for mating: governance is an authority architecture, not a review workflow, and the authority to govern mating events belongs to humans while the labor of performing combination operations is allocable per A1.12.

The framework divides mating activity into two categories:

Mating governance acts — acts requiring human authority. No mating governance act may be delegated to LLMs or to stable cells operating under orchestration rules, though LLMs may draft inputs for human review and approval.

Mating labor acts — acts whose performance is allocable per A1.12. LLMs may perform mating labor acts under human direction, with humans holding authority over the orchestration rules that govern LLM labor.

2.2 Mating governance acts

Four governance acts are required for every mating event across all three patterns:

Mating decision. Humans decide that a mating event should occur. The decision specifies: (a) which entities will participate as parents — whether cells, aspects, or Selves; (b) what the source of each

parent's content is; and (c) which of the three mating patterns governs the combination. The mating decision is the fundamental governance act; no mating event proceeds without it. An LLM cannot make the mating decision; it may present analysis or draft proposals for human review, but the decision itself is a human governance act per A1.01.

Pattern selection. Humans select which mating pattern — Union per B2.46, Selective merge per B2.47, or Lineage-preserved union per B2.48 — is appropriate for the operational context. Pattern selection is architecturally consequential: the chosen pattern determines the combination mechanics, the lineage treatment in the offspring, and — for Selective merge — whether additional governance acts apply. Pattern selection may be made simultaneously with or as a component of the mating decision; it is presented separately here because its architectural consequences are distinct.

Offspring specification approval. Before birth, humans approve the offspring specification produced by the combination operation per B1.09. This approval confirms that the combination result meets the intended design and that the offspring specification is authoritative. Offspring specification approval applies equally across all three patterns: the combination mechanics differ per pattern, but the approval requirement is invariant. Where LLMs draft the combination result for human review, the approval act is the governance boundary between LLM labor and human authority.

Mating authorization. Humans formally authorize that the mating event proceed and that the approved offspring specification is authoritative. Mating authorization is the closing governance act; it is the signal to downstream processes — including offspring birth per B1.06 and mating record creation per §2.3 — that the mating event is governed and complete from the governance side.

Selective merge additional governance act. For the Selective merge pattern per B2.47, an additional governance act is required: selection rule authoring per A2.04. Humans author or formally approve the selection rules that determine which elements from each parent cross into the offspring. The selection rules are substrate content under human authority; they may be drafted by LLMs for human approval, but the authored or approved version is what governs the combination operation. This additional act does not apply to Union or Lineage-preserved union patterns.

2.3 Mating labor acts

Four labor acts are required for mating events. Their performance is allocable per A1.12 — humans may perform them directly, LLMs may perform them under human direction, or stable cells may perform them under orchestration rules where they have reached operational stability.

Combination operations. Executing the DNA combination per the chosen pattern: combining all parent content in Union, applying selection rules in Selective merge, generating the combined specification with embedded lineage pointers in Lineage-preserved union. LLMs may perform combination operations per A1.12 — for Union, generating a unified offspring specification from parent DNA and action layers; for Selective merge, applying the human-authored selection rules to identify which elements from each parent enter the offspring; for Lineage-preserved union, generating a combined specification with parent cell references embedded as substrate content. LLM-drafted combination results require human review and approval at the offspring specification approval

governance act before proceeding.

Selection rule application. For the Selective merge pattern, applying the authored selection rules to the parent content to produce the pre-curated element set for inclusion in the offspring. This is a labor act — application of human-authored rules — distinct from the governance act of authoring or approving those rules. The distinction matters: the same LLM that drafted the combination result under labor allocation cannot author the selection rules under which it operates.

Consistency checking. Verifying that the combination result is consistent with the mating governance decision: that the offspring specification reflects the chosen pattern, that all elements included are consistent with the selection rules (Selective merge) or the union commitment (Union and Lineage-preserved union), and that the result does not introduce material deviations from governance intent that would require a new governance decision. Consistency checking may surface issues for human review; it does not itself constitute the offspring specification approval governance act.

Mating record creation. Creating the A2.40 mating event record that documents the mating event in the substrate. The mating event record is distinct from the offspring birth record (§2.4). Its creation is a labor act; its content is authoritative substrate state once created. The mating record captures: which entities participated as parents (by substrate identifier), which mating pattern was used, when the mating event was authorized, by whom authorization was granted, what offspring specification resulted, and references to the governance acts that authorized the event.

2.4 Mating lineage establishment

Mating establishes lineage for the offspring at the moment of offspring birth. Lineage establishment follows the structure B2.43 specifies for birth lineage, applied at mating scope: the offspring birth record per A2.40 is the lineage anchor.

Offspring birth record as lineage anchor. The offspring's birth record per A2.40 carries provenance fields that anchor lineage. For mating-born offspring, those fields reference both parent lineages, making the offspring's origin traceable through the mating event to both parents. This is architecturally distinct from birth-by-creation (B2.43): creation-born offspring have a single origination record; mating-born offspring have a birth record that fans out to two (or more) lineage ancestors.

Cross-lineage origin. The offspring's lineage tree fans out through both parents. Each parent has its own lineage chain — its own birth record, its own ancestry going back to its origination. The mating-born offspring inherits both chains. The lineage treatment varies by pattern: in Union and Selective merge, parent lineages are referenced in the offspring birth record's provenance fields per A2.40; in Lineage-preserved union per B2.48, parent lineage chains are explicitly embedded as substrate content in the offspring, making every element of the offspring's substrate traceable to its parent source.

Mating event record separate from offspring birth record. The mating event record per §2.3 and the offspring birth record are distinct substrate artifacts. The mating event record captures the mating event itself — the governance acts, the participants, the pattern used, the authorization. The offspring birth record captures the offspring's creation as a governed entity — its specification, its provenance, its

lineage anchors. The offspring birth record references the mating event record in its provenance; this reference is what connects offspring lineage to the mating event and through it to both parents.

Post-mating conflict registry. For Union and Lineage-preserved union patterns, merge-time conflicts per Paper 1 §5 may be captured in a post-mating conflict registry. Merge conflicts are persistent first-class substrate state in the offspring; the conflict registry is the governance artifact that makes them visible and actionable. Humans address conflicts through post-birth directed selection per B1.14; the conflict registry is the substrate object through which that governance operates. Populating the post-mating conflict registry is a mating labor act; the governance decision about how to resolve each conflict is a human authority act addressed post-birth.

2.5 Cross-partner mating per A2.47

When mating involves entities from different composition partners — where “partner” names a composition boundary per A1.13 — cross-partner authority must be established per A2.47 before mating can be governed. Cross-partner authority governs which entities from each partner’s substrate may participate in a mating event, under which patterns, and subject to which governance requirements.

Without established cross-partner authority per A2.47, no mating event spanning partner boundaries can proceed under the CKS governance framework. The governance acts in §2.2 presuppose that the humans exercising them have authority over both parent entities; where parents come from different partner substrates, that authority must be established at the composition level before mating-level governance can operate.

Cross-partner mating governance does not differ from single-partner mating governance in its act structure: the same four governance acts apply. The additional requirement is that the mating decision in §2.2 must include establishment of or reference to the A2.47 cross-partner authority under which the decision is made.

3. What makes mating governance and lineage establishment architecturally distinctive

Conventional AI component combination — merging agent configurations, composing prompt templates, chaining system components — is typically automatic or ad-hoc. Components are merged by engineers following application logic; no explicit governance framework governs the combination decision, no pattern selection is formalized, no offspring specification approval is required, and no lineage anchoring connects the combined artifact to both sources. The combination happens; it is not governed.

CKS mating governance differs on every one of these axes. The governance framework is explicit: four governance acts, each performed by humans per A1.01, covering the decision to mate, the pattern choice, the offspring specification review, and the final authorization. The framework is consistent: the same four acts apply regardless of which mating pattern is chosen. The lineage establishment is architecturally formal: the offspring birth record anchors lineage, references the mating event record, and fans out through both parent lineages. The treatment of merge-time conflicts is architecturally explicit for Union and Lineage-preserved union: conflicts are first-class substrate state, not discarded or silently resolved.

The consistency across patterns is itself architecturally significant. Union, Selective merge, and Lineage-preserved union differ substantially in their combination mechanics and lineage treatment. A governance framework that is consistent across all three is not a by-product of the pattern designs; it is a designed architectural property. The framework's consistency is what makes it possible to reason about mating governance without reasoning separately about each pattern — the pattern determines the combination mechanics; the governance framework is invariant.

4. Inherited Paper 1 commitments

The mating governance and lineage establishment framework inherits from Paper 1 without redefense. The most load-bearing inheritances are as follows.

A1.01 (human-governed). All four mating governance acts — mating decision, pattern selection, offspring specification approval, mating authorization — are human governance acts per A1.01. No mating governance act may be delegated to LLMs or automation. The governance framework applies this commitment at mating scope across all three patterns.

A1.12 (labor allocation). Mating labor acts — combination operations, selection rule application, consistency checking, mating record creation — are allocable per A1.12. LLMs may perform them under human direction. Where LLMs draft combination results, the offspring specification approval governance act is the boundary at which human authority over LLM labor is exercised.

A2.04 (rule authoring). For Selective merge, selection rules are substrate content authored by humans per A2.04. The selection rule authoring governance act is the application of A2.04's rule authoring commitment to the mating context. Selection rules authored under A2.04 are authoritative substrate content; they govern LLM or human labor in applying the selection to parent content.

A2.40 (six provenance metadata fields). Mating event records and offspring birth records are substrate artifacts that satisfy A2.40's six provenance metadata requirements. The mating event record's six fields capture: what the mating event was, who the participants were, when it occurred, by whom it was authorized, what the resulting offspring specification is, and what governance acts preceded it. The offspring birth record's provenance fields reference the mating event record, instantiating A2.40's requirement that substrate content carry traceable provenance.

A1.07 (path retraceability). Mating lineage is the application of A1.07's path retraceability commitment to mating scope. The offspring birth record references the mating event record; the mating event record references both parent entities; each parent entity has its own lineage chain per B2.43. The full lineage path from offspring to both parents' origins is retraceable through substrate records without external reconstruction.

A1.13 (composition requirements). Offspring entities produced by mating must satisfy A1.13's composition requirements. Per-substrate human governance preservation (A2.66), conflict preservation across boundaries (A2.67), and the other composition requirements apply to mating-born offspring as they apply to any CKS entity. Cross-partner mating additionally requires A2.47 authority distribution as a precondition for cross-boundary governance.

5. Cross-pattern consistency and Selective merge's additional governance act

The governance framework is consistent across all three mating patterns in the following sense: every mating event, regardless of pattern, involves exactly the four governance acts in §2.2, the four labor acts in §2.3, and the lineage establishment mechanism in §2.4. The pattern choice does not alter this structure.

What varies across patterns is the combination mechanics (covered in B2.46, B2.47, B2.48) and the lineage treatment in the offspring. Specifically:

Union and Lineage-preserved union produce offspring that carry all parent content (with conflict preservation for Union). Their lineage treatment differs: Union references parent lineages in the offspring birth record's provenance fields; Lineage-preserved union embeds parent lineage chains as substrate content in the offspring itself. Both satisfy the lineage establishment framework of §2.4; they satisfy it differently.

Selective merge produces offspring whose content footprint is curated to the human-authored selection. Its lineage treatment references parent lineages in the offspring birth record's provenance fields, as Union does. Its governance structure differs from Union and Lineage-preserved union in one respect: the selection rule authoring governance act applies, making Selective merge's governance sequence five acts rather than four.

The five-act structure for Selective merge is not an exception to the framework; it is the framework applied to a pattern that requires an additional governance act by design. The selection rules are the mechanism by which humans govern which elements enter the offspring from each parent; without human authority over the rules, the Selective merge pattern would reduce to automated curation, which the CKS architecture does not commit to.

6. Operational implications

Configure mating governance workflows per operational requirements. The mating governance framework specifies the governance acts required, not the workflow through which they are executed. Deployments configure workflows per their operational requirements: the order in which governance acts are completed, the tools used to record them, the humans authorized to perform each act, and the confirmation mechanisms that link governance acts to downstream mating labor. Standard mating governance workflow templates for each pattern are operationally useful; they are not architectural commitments.

LLM drafting with human review. LLMs may draft combination operations — proposing Union offspring specifications, applying Selective merge selection rules, generating Lineage-preserved union specifications with parent pointers — for human review before the offspring specification approval governance act. The architecture supports this allocation per A1.12; it requires that LLM-drafted results be reviewed and approved by humans before mating authorization proceeds, not that LLMs be excluded from the combination process.

Mating governance events recorded with full provenance. Mating event records per A2.40 are authoritative substrate records. Their creation is a labor act; their existence and completeness is a governance concern. A mating event without a complete mating event record is a governance gap: the lineage chain from offspring to both parents cannot be reconstructed from the offspring birth record alone if the mating event record is absent or incomplete.

Cross-partner mating requires A2.47 authority establishment first. Operational deployments that anticipate cross-partner mating must establish A2.47 cross-partner authority before any cross-partner mating governance decision can be made. This is not a workflow recommendation; it is a precondition. Attempting to govern a cross-partner mating event without established cross-partner authority means that at least one parent entity is not under the governance authority of the humans making the mating decision.

Post-mating conflict registry for Union and Lineage-preserved union. Deployments using Union or Lineage-preserved union patterns should establish a post-mating conflict registry as part of offspring birth. The registry makes merge-time conflicts visible and actionable for human governance through post-birth directed selection per B1.14. The registry’s population is a mating labor act; its governance is a post-birth human authority responsibility.

Offspring specification review before birth per B2.44. The offspring specification approval governance act in §2.2 is the link between mating governance and birth governance. No mating-born offspring proceeds to birth per B1.06 without an approved offspring specification. The review covers both the content of the combination result and the completeness of the mating governance record.

7. Limits of mating governance and lineage establishment

Does not prescribe specific governance workflows. The framework specifies the governance acts and their required sequence (mating decision precedes pattern selection; pattern selection precedes combination operations; combination operations precede offspring specification approval; approval precedes authorization). It does not prescribe how those acts are executed, recorded, or linked in any particular deployment.

Governance is consistent, not uniform. “Consistent” means the same governance structure applies regardless of pattern choice. “Uniform” would mean the governance acts are identical for all patterns. They are not identical: Selective merge requires selection rule authoring as an additional governance act. Consistent means Union, Selective merge, and Lineage-preserved union all operate within the same governance framework; it does not mean they have identical governance act counts.

Mating governance is not mating labor. The governance vs. labor distinction per B2.41 must be maintained. Governance acts (§2.2) require human authority and cannot be delegated. Labor acts (§2.3) are allocable and may be performed by LLMs under human direction. The combination operations that LLMs perform are labor; the mating decision that authorizes those operations is governance. Conflating them — treating combination operations as governance acts or treating mating decisions as delegable to LLMs — misreads the architecture.

Mating lineage does not replace post-birth operations lineage. The mating lineage established at birth is the lineage anchor. Every post-birth operation on the offspring — every action taken, every DNA evolution applied, every directed selection — extends the lineage chain forward from the birth anchor. Mating lineage is the foundation of the offspring’s full lineage chain; it is not the complete chain. B2.43’s birth lineage establishment applies to mating-born offspring as to creation-born offspring; the difference is that the birth anchor fans out through two parents rather than one.

Does not prescribe specific inter-partner governance arrangements. Cross-partner mating requires A2.47 authority distribution as a precondition. How inter-partner authority is established — through contractual arrangements, through shared governance structures, through explicit mating authorization protocols — is not specified by the mating governance framework. The framework requires that cross-partner authority exist and be referenced in the mating decision; it does not prescribe the arrangements through which that authority is created.

Mating governance closes the three-pattern governance account. B2.49 formalizes the governance framework common to all three patterns. B2.50 will formalize mating verification — the confirmation that governance acts were properly executed before mating-born offspring are admitted to the operational substrate. B2.49 does not extend into mating verification; that formalization belongs to B2.50.

8. One-sentence architectural test

A CKS mating event is governed if and only if humans have exercised the mating decision, pattern selection, offspring specification approval, and mating authorization governance acts, with mating labor acts performed under human direction per A1.12, with the mating event and offspring birth separately recorded as A2.40 substrate artifacts, and with the offspring birth record referencing the mating event record as the lineage anchor connecting the offspring to both parent lineages.

9. Why naming as standalone matters and its position in the Phase B2 progression

The mating governance and lineage establishment framework is not derivable from any single mating pattern account. B2.46 formalizes how Union combines parent content and preserves conflicts; it does not formalize the governance acts that authorize a Union event or how Union lineage is anchored in the offspring birth record relative to the mating event record. B2.47 formalizes how Selective merge curates parent content under human-authored rules; it does not formalize the governance acts common to Selective merge and the other two patterns. B2.48 formalizes how Lineage-preserved union embeds parent lineage chains; it does not formalize what governance structure the lineage-preservation commitment operates within.

B2.49 names the framework that makes each pattern account a governed architectural primitive rather than a combination mechanism operating without authority structure. Naming the framework as standalone is what creates a prior-art record of mating governance as an independent formalized commitment — not as an implementation detail of one pattern, not as an implication of birth governance, but as a consistent cross-pattern governance framework with specific acts, a specific governance-vs-labor structure, and a specific lineage anchoring mechanism.

In the Phase B2 progression, B2.49 is the fifth of six notes decomposing B1.10:

B2.45: Mating mechanism specification — the mating primitive and the three-channel collapse

B2.46: Union mating pattern — the full-union combination mechanics and conflict preservation

B2.47: Selective merge mating pattern — the curation mechanics and selection rule authority

B2.48: Lineage-preserved union mating pattern — the embedded-lineage mechanics and provenance

B2.49 (this note): Mating governance and lineage establishment — the consistent cross-pattern governance framework

B2.50 (next): Mating verification — confirming governance completeness before offspring birth

Following B2.50, Phase B2 notes will decompose B1.11 (death) across approximately five notes (B2.51–B2.55) covering death governance, functional obsolescence, lineage supersession, archived addressability, and death-type-specific governance processes. The full Phase B2 progression continues toward the approximately 110-note completion of the operational variant and decomposition record for Paper 2.

Cross-references

Directly load-bearing: B1.10 (mating as cross-layer combination); B2.41 (birth governance vs. labor); B2.43 (birth lineage establishment); B2.45 (mating mechanism specification); B2.46 (Union); B2.47 (Selective merge); B2.48 (Lineage-preserved union); A1.01 (human-governed); A1.12 (labor allocation); A2.04 (rule authoring); A2.40 (six provenance metadata fields); A1.07 (path retraceability); A1.13 (composition requirements); A2.47 (authority distribution).

Directly relevant: B1.06 (birth as governed origination); B1.09 (offspring specification before birth); B1.14 (directed selection); A2.66 (composition requirement: per-substrate human governance); A2.67 (composition requirement: conflict preservation across boundaries).

Subsequent note: B2.50 (mating verification — closes B1.10 decomposition).